

DESPLEGAMENT D'APLICACIONS WEB

Práctica 11

DNS (BIND9)

ALUMNO

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Nota sobre el entorno

El servidor DNS (BIND9) se instala sobre la VM **Ubuntu Server 24.04** (192.168.1.178 , red 192.168.1.0/24). El cliente es la máquina **host con Arch/CachyOS** (no Windows), por lo que los comandos de Windows del enunciado se sustituyen por sus equivalentes de **systemd-resolved**:

Windows	Linux (host, systemd-resolved)
ipconfig /displaydns	resolvectl statistics
ipconfig /flushdns	sudo resolvectl flush-caches
Cambiar DNS del adaptador	sudo resolvectl dns wlan0 <ip>

Como la IP no es la del ejemplo (192.168.5.x) sino 192.168.1.178, la zona inversa es 1.168.192.in-addr.arpa y el PTR es el octeto 178. En la VM, la resolución vía BIND se fija con `resolvectl dns enp0s3 127.0.0.1`. Para comprobar Internet en la VM se usa `ping -4` (la VM no tiene ruta IPv6).

Se actualiza el sistema y se instala **bind9** (en Ubuntu 24.04 el servicio es `named`):

```
sudo apt update && sudo apt upgrade -y
sudo apt install -y bind9
```

[illegible]

Update e instalación

Se verifica que el servicio queda habilitado en el arranque y activo:

```
sudo systemctl is-enabled named
sudo systemctl status named
```

```
castor909@mus:~$ sudo systemctl is-enabled named
enabled
castor909@mus:~$ sudo systemctl status named --no-pager | head -6
● named.service - BIND Domain Name Server
   Loaded: loaded (/usr/lib/systemd/system/named.service; enabled; preset: enabled)
   Active: active (running) since Sun 2026-06-14 17:26:33 UTC; 49s ago
     Docs: man:named(8)
    Main PID: 45001 (named)
      Status: "running"
```

Servicio habilitado y activo

Se abren SSH y BIND9 (puerto 53) en el firewall y se activa:

```
sudo ufw allow ssh
sudo ufw allow bind9
sudo ufw enable
sudo ufw status verbose
```

```
castor909@mus:~$ sudo ufw allow ssh
castor909@mus:~$ sudo ufw allow bind9
castor909@mus:~$ sudo ufw enable
castor909@mus:~$ sudo ufw status verbose
Skipping adding existing rule
Skipping adding existing rule (v6)
Rule added
Rule added (v6)
Command may disrupt existing ssh connections. Proceed with operation (y|n)? y
Firewall is active and enabled on system startup
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), deny (routed)
New profiles: skip

To Action From
--
22/tcp ALLOW IN Anywhere
389/tcp ALLOW IN Anywhere
80/tcp ALLOW IN Anywhere
22/tcp (OpenSSH) ALLOW IN Anywhere
3000/tcp ALLOW IN Anywhere
2222/tcp ALLOW IN Anywhere
21/tcp ALLOW IN Anywhere
20/tcp ALLOW IN Anywhere
4500:6000/tcp ALLOW IN Anywhere
53 (Bind9) ALLOW IN Anywhere
22/tcp (v6) ALLOW IN Anywhere (v6)
389/tcp (v6) ALLOW IN Anywhere (v6)
80/tcp (v6) ALLOW IN Anywhere (v6)
22/tcp (OpenSSH (v6)) ALLOW IN Anywhere (v6)
3000/tcp (v6) ALLOW IN Anywhere (v6)
2222/tcp (v6) ALLOW IN Anywhere (v6)
21/tcp (v6) ALLOW IN Anywhere (v6)
20/tcp (v6) ALLOW IN Anywhere (v6)
4500:6000/tcp (v6) ALLOW IN Anywhere (v6)
53 (Bind9 (v6)) ALLOW IN Anywhere (v6)
```

ufw con ssh y bind9

2. Configuración de `named.conf.options`

Se editan `/etc/bind/named.conf.options` (`forwarders`, `allow-query` con la red/máscara del servidor, `listen-on` IPv4 y `dnssec-validation no`) y `/etc/default/named` (`OPTIONS="-u bind -4"` para que BIND resuelva solo IPv4):

```
castor909@mus:~$ cat /etc/bind/named.conf.options
cat /etc/default/named
options {
    directory "/var/cache/bind";

    forwarders {
        8.8.8.8;
        8.8.4.4;
    };

    allow-query { localhost; 192.168.1.0/24; };
    listen-on { 127.0.0.1; 192.168.1.178; };

    dnssec-validation no;

    recursion yes;
};
#
# run resolvconf?
RESOLVCONF=no
# startup options for the server
OPTIONS="-u bind -4"
```

`named.conf.options` y `/etc/default/named`

Se verifica la sintaxis, se reinicia y se comprueba el estado:

```
sudo named-checkconf
sudo systemctl restart named
sudo systemctl status named
```

```
castor909@mus:~$ sudo named-checkconf
sudo systemctl restart named
sudo systemctl status named --no-pager | head -6
● named.service - BIND Domain Name Server
   Loaded: loaded (/usr/lib/systemd/system/named.service; enabled; preset: enabled)
   Active: active (running) since Sun 2026-06-14 17:33:03 UTC; 13ms ago
     Docs: man:named(8)
    Main PID: 46171 (named)
      Status: "running"
```

`named-checkconf` y `status`

3. Zonas directa e inversa

En `/etc/bind/named.conf.local` se declaran la zona `dawserver.com` y la inversa `1.168.192.in-addr.arpa`, y en `/etc/bind/zonas/` se crean los ficheros `db.dawserver.com` (hostname `dawser`, `www` y `@` → `192.168.1.178`) y `db.192.168.1` (PTR `178` → `dawser.dawserver.com`):

```

castor909@mus:~$ cat /etc/bind/named.conf.local
cat /etc/bind/zonas/db.dawserver.com
cat /etc/bind/zonas/db.192.168.1
//
// Do any local configuration here
//

// Consider adding the 1918 zones here, if they are not used in your
// organization
//include "/etc/bind/zones.rfc1918";

zone "dawserver.com" IN {
    type master;
    file "/etc/bind/zonas/db.dawserver.com";
};

zone "1.168.192.in-addr.arpa" {
    type master;
    file "/etc/bind/zonas/db.192.168.1";
};
$TTL 1D
@ IN SOA dawser.dawserver.com. admin.dawserver.com. (
    2026061401 ; Serial
    12h        ; Refresh
    15m        ; Retry
    3w         ; Expire
    2h )       ; Negative Cache TTL

@ IN NS  dawser.dawserver.com.
dawser IN A  192.168.1.178
www IN A  192.168.1.178
@ IN A  192.168.1.178
$TTL 1D
@ IN SOA dawser.dawserver.com. admin.dawserver.com. (
    2026061401 ; Serial
    12h        ; Refresh
    15m        ; Retry
    3w         ; Expire
    2h )       ; Negative Cache TTL

@ IN NS  dawser.dawserver.com.
178 IN PTR dawser.dawserver.com.

```

named.conf.local y ficheros de zona

Se validan ambas zonas:

```

sudo named-checkzone dawserver.com /etc/bind/zonas/db.dawserver.com
sudo named-checkzone 1.168.192.in-addr.arpa /etc/bind/zonas/db.192.168.1

```

```

castor909@mus:~$ sudo named-checkzone dawserver.com /etc/bind/zonas/db.dawserver.com
sudo named-checkzone 1.168.192.in-addr.arpa /etc/bind/zonas/db.192.168.1
zone dawserver.com/IN: loaded serial 2026061401
OK
zone 1.168.192.in-addr.arpa/IN: loaded serial 2026061401
OK

```

named-checkzone OK

Tras reiniciar, el log muestra ambas zonas cargadas (loaded serial):

Zonas cargadas

Se comprueba que la VM tiene Internet (ping -4 google.com):

Ping a google desde la VM

Host sin resolución

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```

castor909@mus:~$ sudo cp /etc/hosts /etc/hosts.bak
printf '127.0.0.1 localhost\n' | sudo tee /etc/hosts
sudo resolvectl dns enp0s3 127.0.0.1
sudo resolvectl domain enp0s3 '~.'
resolvectl status enp0s3 | grep -i "DNS Server"
ping -c2 www.dawserver.com
ping -c2 google.com
127.0.0.1 localhost
Current DNS Server: 127.0.0.1
DNS Servers: 127.0.0.1
PING www.dawserver.com (192.168.1.178) 56(84) bytes of data.
64 bytes from dawser.dawserver.com (192.168.1.178): icmp_seq=1 ttl=64 time=0.034 ms
64 bytes from dawser.dawserver.com (192.168.1.178): icmp_seq=2 ttl=64 time=0.069 ms

--- www.dawserver.com ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1041ms
rtt min/avg/max/mdev = 0.034/0.051/0.069/0.017 ms
PING google.com (2a00:1450:4003:80b::200e) 56 data bytes

--- google.com ping statistics ---
2 packets transmitted, 0 received, 100% packet loss, time 1006ms

```

VM resuelve por su DNS

Se deshabilita el firewall de la VM (paso requerido):

```
sudo ufw disable
```

```

castor909@mus:~$ sudo ufw disable
sudo ufw status
Firewall stopped and disabled on system startup
Status: inactive

```

ufw disable

Se configura el sitio Apache con `ServerName www.dawserver.com`. En esta VM el puerto 80 lo servía nginx y no existía `sitio1.conf`, por lo que se crea `/etc/apache2/sites-available/sitio1.conf`, se habilita y se conmuta el puerto 80 a Apache:

```

castor909@mus:~$ sudo mkdir -p /var/www/sitio1
echo '<!--sitio1 - www.dawserver.com (Apache + BIND)-->' | sudo tee /var/www/sitio1/index.html
sudo tee /etc/apache2/sites-available/sitio1.conf > /dev/null << 'EOF'
<VirtualHost >000
    ServerName www.dawserver.com
    ServerAlias dawserver.com
    DocumentRoot /var/www/sitio1
</VirtualHost>
EOF
sudo a2ensite sitio1
sudo a2dissite 000-default
sudo systemctl stop nginx
sudo systemctl disable nginx
sudo systemctl enable apache2
sudo systemctl start apache2
sudo apache2ctl configtest
curl http://www.dawserver.com/
<!--sitio1 - www.dawserver.com (Apache + BIND)--></h1>
Enabling site sitio1.
To activate the new configuration, you need to run:
systemctl reload apache2
Site 000-default disabled.
To activate the new configuration, you need to run:
systemctl reload apache2
Synchronizing state of nginx.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install disable nginx
Removed /etc/systemd/system/multi-user.target.wants/nginx.service.
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2
Created symlink /etc/systemd/system/multi-user.target.wants/apache2.service - /usr/lib/systemd/system/apache2.service.
AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 2a0c:5a86:d00a:7000:a00:27ff:fed0:7e81. Set the 'ServerName' directive globally to suppress this message
Syntax OK
<!--sitio1 - www.dawserver.com (Apache + BIND)--></h1>

```

sitio1.conf y Apache activo

5. El host usa el DNS creado (captura DNS)

En el host se apunta el DNS al servidor de la VM y se vacía la caché (equivalentes de `ipconfig /displaydns` y `/flushdns`):

```
resolvectl statistics
sudo resolvectl flush-caches
sudo resolvectl dns wlan0 192.168.1.178
resolvectl query www.dawserver.com
```

```
castor909@mus:~$ resolvectl statistics | head -12
sudo resolvectl flush-caches
sudo resolvectl dns wlan0 192.168.1.178
sudo resolvectl domain wlan0 '~.'
resolvectl status wlan0 | grep -iE "DNS Server|Current DNS"
resolvectl query www.dawserver.com
Failed to connect to query monitoring service /run/systemd/resolve/io.systemd.Resolve.Monitor: Permission denied
Failed to resolve interface "wlan0": No such device
Failed to resolve interface "wlan0": No such device
Failed to resolve interface "wlan0", ignoring: No such device
www.dawserver.com: 192.168.1.178

-- Information acquired via protocol DNS in 3.0ms.
-- Data is authenticated: no; Data was acquired via local or encrypted transport: no
-- Data from: network
resolvectl statistics | head -12
sudo resolvectl flush-caches
sudo resolvectl dns wlan0 192.168.1.178
sudo resolvectl domain wlan0 '~.'
resolvectl status wlan0 | grep -iE "DNS Server|Current DNS"
resolvectl query www.dawserver.com

fish: Expected a variable name after this $.
castor909@mus:~$ resolvectl statistics | head -12
^
resolvectl statistics | head -12
sudo resolvectl flush-caches
sudo resolvectl dns wlan0 192.168.1.178
sudo resolvectl domain wlan0 '~.'
resolvectl status wlan0 | grep -iE "DNS Server|Current DNS"
resolvectl query www.dawserver.com

Transactions
Current Transactions: 0
Total Transactions: 10865

Cache
Current Cache Size: 151
Cache Hits: 2492
Cache Misses: 7390

Failure Transactions
Total Timeouts: 107
Total Timeouts (Stale Data Served): 0
[sudo] password for castor909:
Current DNS Server: 192.168.1.178
DNS Servers: 192.168.1.178
www.dawserver.com: 192.168.1.178 -- link: wlan0

-- Information acquired via protocol DNS in 773us.
-- Data is authenticated: no; Data was acquired via local or encrypted transport: no
-- Data from: network
8s
```

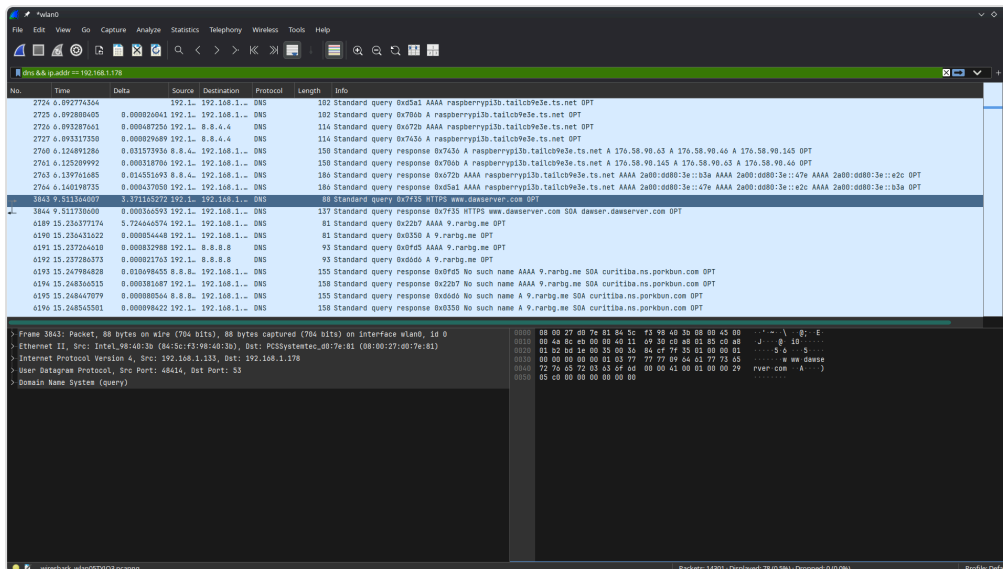
Host apuntando al DNS de la VM

Con Wireshark capturando, se accede a `http://www.dawserver.com` desde el navegador, mostrando el sitio `sitio1`:



Navegador en `www.dawserver.com`

La captura (filtro `dns && ip.addr == 192.168.1.178`) confirma que la consulta DNS de `www.dawserver.com` se realiza al servidor de la VM:



Captura DNS hacia la VM

BONUS — DNS primario = VM, secundario = 8.8.8.8

Se configura el host con el DNS de la VM como primario y 8.8.8.8 como secundario:

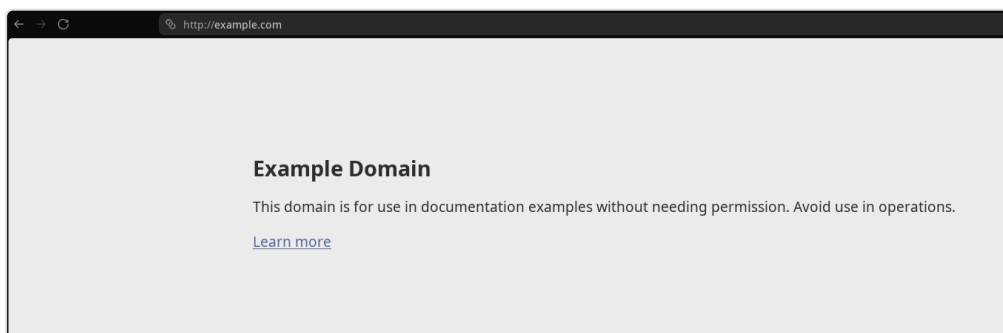
```
sudo resolvectl dns wlan0 192.168.1.178 8.8.8.8
sudo resolvectl flush-caches
resolvectl status wlan0
```

```
> sudo resolvectl dns wlan0 192.168.1.178 8.8.8.8
sudo resolvectl flush-caches
resolvectl status wlan0 | grep -iE "DNS Server|Current DNS"

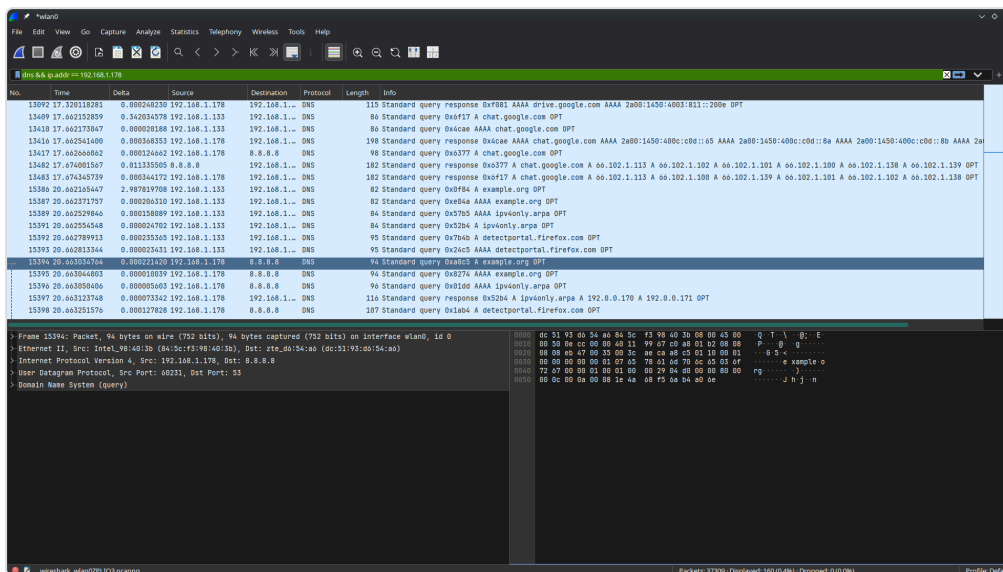
[sudo] password for castor909:
Current DNS Server: 192.168.1.178
DNS Servers: 192.168.1.178 8.8.8.8
```

DNS primario VM + secundario Google

Se navega a un sitio **distinto** de dawserver.com (example.com):



La captura muestra que incluso para un dominio externo, la consulta DNS se envía **al servidor de la VM** (192.168.1.178), que la reenvía a sus forwarders:



DNS de sitio externo hacia la VM

Conclusión

Se ha montado un servidor **DNS autoritativo con BIND9** para el dominio **dawserver.com**, con zona directa e inversa, forwarders y resolución IPv4. Se ha comprobado que tanto la VM como el host resuelven **www.dawserver.com** a través de este servidor, accediendo al sitio Apache asociado, y que el host puede usar la VM como servidor DNS primario para toda su navegación.